

# None of the four recipe knobs moves on-policy bystander marker leakage

Held-out bystander subset (n=15 personas, 20 generic eval questions, 1 seed). Threshold for declaring a knob-effect was  $\geq 1.0$  nat with a permutation  $p < 0.10$ ; observed cross-knob range is  $\sim 0.003$  nats, three orders of magnitude smaller.

